



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

FULL TEST

05-01-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS, TAAS, TEAS, TIAS, TMAS, I(A), LIVE-I(A), TRAS, I, LIVE-I & III] (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	B	B	D	A	B	C	C	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	C	B	B	D	A	B	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4	2	3	40	16	27	8	8	60	6

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	B	B	B	A	B	B	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	B	D	C	C	A	C	B	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	8	3	3	75	1200	6	4	8	4	3

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	B	A	C	D	B	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	D	D	C	D	B	B	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	112	75	2	10	21	4	3	2	362	9

HINT – SHEET

PART-1 : PHYSICS

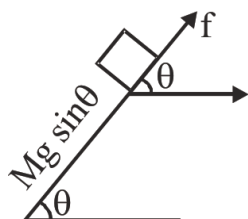
SECTION-I

1. Ans (B)

If block is at rest

$$f = mg \sin \theta$$

$$\therefore f_{\text{horizontal}} = mg \sin \theta \cos \theta = \frac{mg}{2} (\sin 2\theta)$$



$$\therefore f_{\text{max horizontal}} \text{ is } \frac{mg}{2}$$

2. Ans (A)

$$\frac{1}{2} kx^2 = \frac{1}{2} mv^2$$

$$\frac{1}{2} k(x')^2 = \frac{1}{2} 4m(3v)^2$$

$$\left(\frac{x'}{x}\right)^2 = 36$$

$$x' = 6x$$

3. **Ans (B)**

At the time when it just starts to slip

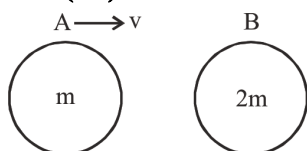
$$f_{\max} = m \sqrt{\left(\frac{v_0^2}{R}\right)^2 + a_0^2}$$

$$\mu mg = m \sqrt{\frac{v_0^4}{R^2} + a_0^2}$$

$$\mu^2 g^2 = \frac{v_0^4}{R^2} + a_0^2$$

$$R^2 = \frac{v_0^4}{\mu^2 g^2 - a_0^2} \Rightarrow R = \frac{v_0^2}{\sqrt{\mu^2 g^2 - a_0^2}}$$

4. **Ans (B)**



For maximum loss in kinetic energy of 'A'

Final velocity of 'A' = 0

$$\Rightarrow P_i = P_f$$

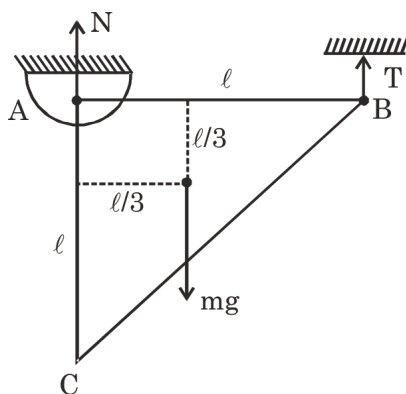
$$mv = 0 + 2mv'$$

$$v' = \frac{v}{2}$$

$$e = \frac{\frac{v}{2} - 0}{\frac{v}{2} - 0} = \frac{1}{2} = 0.5$$

5. **Ans (D)**

FBD



$$\tau_B = 0$$

$$m\ell = mg \frac{2\ell}{3}$$

$$\Rightarrow N = \frac{2mg}{3}$$

6. **Ans (A)**

$$v_{\max} = a\omega = 3 \times 10 = 30$$

7. **Ans (B)**

$$Q = nC_m \Delta T \quad \dots(1)$$

$$\Delta u = \frac{Q}{2} = \frac{F}{2} n R \Delta T \quad \dots(2)$$

$$2 = \frac{C_m}{\frac{F}{2} R} \Rightarrow C_m = F.R = 3R$$

8. **Ans (C)**

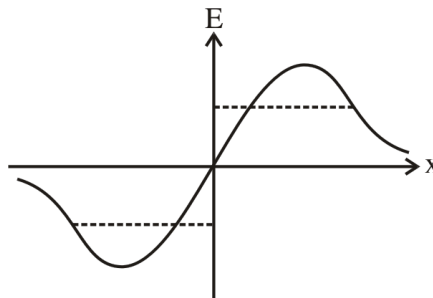
In the first case, the temperature difference between container and surrounding is greater as compared to the second case. So, the rate of loss of heat is quicker.

9. **Ans (C)**

$$mg = 600 \times 0.6g + 400 \times 1g$$

$$m = 760$$

10. **Ans (D)**



11. **Ans (D)**

$$i_g r_g = i_s r_s$$

$$i_s \approx i_g + i_s$$

$$r_s = \frac{i_g r_g}{i_s} = \frac{i r_g}{i_s} = 10^{-3}$$

12. **Ans (C)**

$$E = B4rv + B2r2v$$

13. **Ans (C)**

$$\frac{1}{x_i} - \frac{1}{-30} = \frac{1}{20}$$

$$x_i = 60$$

$$\frac{\ell_i}{\ell_o} = \frac{x_i}{x_o}$$

$$\ell_i = 8 \text{ cm}$$

14. **Ans (B)**

$$\frac{I}{2} \cos^2 \theta \cos^2 \theta = \frac{I}{3}$$

$$\cos^4 \theta = \frac{2}{3}$$

$$\cos \theta = \left(\frac{2}{3}\right)^{1/4}$$

15. **Ans (B)**

We know velocity of electron in nth shell of hydrogen atom is given by

$$v = \frac{2\pi kZe^2}{nh}$$

$$\therefore v \propto \frac{1}{n}$$

16. **Ans (D)**

$$\vec{P}_1 = \frac{h}{\lambda_1} \hat{i} \quad \vec{P}_2 = \frac{h}{\lambda_2} \hat{j}$$

$$\vec{P}_1 = \frac{h}{\lambda_1} \hat{i} \quad \& \quad \vec{P}_2 = \frac{h}{\lambda_2} \hat{j}$$

Using momentum conservation

$$\vec{P} = \vec{P}_1 + \vec{P}_2$$

$$= \frac{h}{\lambda_1} \hat{i} + \frac{h}{\lambda_2} \hat{j}$$

$$|\vec{P}| = \sqrt{\left(\frac{h}{\lambda_1}\right)^2 + \left(\frac{h}{\lambda_2}\right)^2}$$

$$\frac{h}{\lambda} = \sqrt{\left(\frac{h}{\lambda_1}\right)^2 + \left(\frac{h}{\lambda_2}\right)^2}$$

$$\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$$

17. **Ans (A)**

$$B.E. = [\Delta m] \cdot c^2$$

$$M_{\text{expected}} = ZM_p + (A - Z)M_n$$

$$= 50 [1.00783] + 70 [1.00867]$$

$$M_{\text{actual}} = 119.902199$$

$$B.E. = [50 [1.00783] + 70 [1.00867] - 119.902199]$$

$$\times 931$$

$$= 1020.56$$

$$\frac{BE}{\text{nucleon}} = \frac{1020.56}{120} = 8.5 \text{ MeV}$$

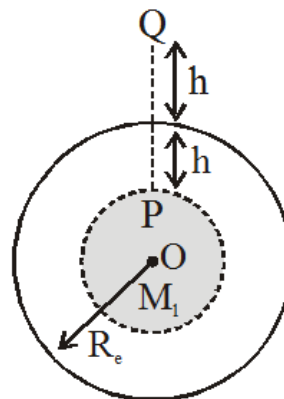
18. **Ans (B)**

$$i_1 = i_2 + i_z$$

$$\frac{15 - 10}{500} = \frac{10 - 0}{1500} + i_z$$

$$i_z = 3.3 \text{ mA}$$

19. **Ans (A)**



♦ M = mass of earth

M₁ = mass of shaded portion

R = Radius of earth

$$\diamond M_1 = \frac{M}{\frac{4}{3}\pi R^3} \cdot \frac{4}{3}\pi (R-h)^3$$

$$= \frac{M(R-h)^3}{R^3}$$

♦ Weight of body is same at P and Q

$$\text{i.e. } mg_P = mg_Q$$

$$g_P = g_Q$$

$$\frac{GM_1}{(R-h)^2} = \frac{GM}{(R+h)^2}$$

$$\frac{GM(R-h)^3}{(R-h)^2 R^3} = \frac{GM}{(R+h)^2}$$

$$(R-h)(R+h)^2 = R^3$$

$$R^3 - hR^2 - h^2R - h^3 + 2R^2h - 2Rh^2 = R^3$$

$$R^2 - Rh^2 - h^3 = 0$$

$$R^2 - Rh - h^2 = 0$$

$$h^2 + Rh - R^2 = 0 \Rightarrow h = \frac{-R \pm \sqrt{R^2 + 4R^2}}{2}$$

$$\text{ie } h = \frac{-R + \sqrt{5}R}{2} = \left(\frac{\sqrt{5}-1}{2}\right)R$$

20. **Ans (B)**

$$A_{\text{res}}^2 = A^2 + A^2 + 2AA \cos 90^\circ$$

$$A_{\text{res}} = \sqrt{2}A$$

$$E = \frac{1}{2} m \omega^2 (\sqrt{2}A)^2$$

$$= m \omega^2 A^2$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (4)**

$$P = \frac{V^2}{Z^2} R = \frac{V^2 \times 400}{\left(\left(50\pi^2 - \frac{10^6}{50\pi \times \pi \times 10} \right)^2 + 400^2 \right)}$$

$$P = \frac{V^2 \times 400}{(500 - 200)^2 + 400^2} = \frac{V^2 \times 400}{500 \times 500} = \frac{4 \times 10}{25 \times 10} = 4W$$

2. **Ans (2)**

$$i = \frac{10}{3+2} = 2A$$

$$i_1 = 1A$$

$$i_3 = 1A$$

$$pd = 3 \times 1 - 1 \times 1 = 2$$

3. **Ans (3)**

$$V = 6a^3$$

$$\ln V = \ln 6 + 3 \ln a$$

$$\frac{dV}{V} = 0 + \frac{3da}{a}$$

$$\frac{dV}{V} \times 100 = \frac{3da}{a} \times 100 = 3\%$$

4. **Ans (40)**

Maximum amplitude

$$A_{\max} = A_m + A_C$$

$$\Rightarrow V_{\max} = V_m + V_C$$

$$200 = V_m + 160$$

$$V_m = 40$$

$$\therefore \text{Peak voltage } A_m = 40$$

Ans. 40

5. **Ans (16)**

$$P = \frac{dq}{dt} = \sigma A e T^4 = \sigma \times 4\pi R^2 \times e \times T^4$$

$$\text{By, volume conservation, } 64 \times \frac{4\pi}{3} R^3 = \frac{4}{3} \pi R_1^3$$

$$\Rightarrow R_1 = 4R$$

$$np = \sigma \times 4\pi (R_1)^2 \times e \times T^4 = \sigma \times 4\pi \times e \times T^4 \times 16 \times R^2$$

$$= 16P$$

$$\therefore n = 16$$

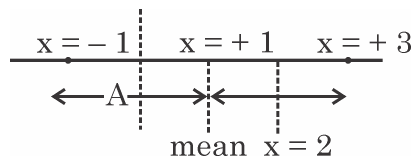
6. **Ans (27)**

$$\frac{1}{\lambda_\alpha} = \frac{3R}{4} (Z-1)^2$$

$$\Rightarrow (Z-1) = \sqrt{\frac{4}{3R\lambda_\alpha}} = \sqrt{\frac{4}{3 \times 1.1 \times 10^7 \times 1.8 \times 10^{-10}}}$$

$$= \frac{200}{3} \sqrt{\frac{5}{33}} = \frac{78}{3} = 26 \Rightarrow Z = 27$$

7. **Ans (8)**



$$\frac{T}{4} = 3 \text{ sec.}$$

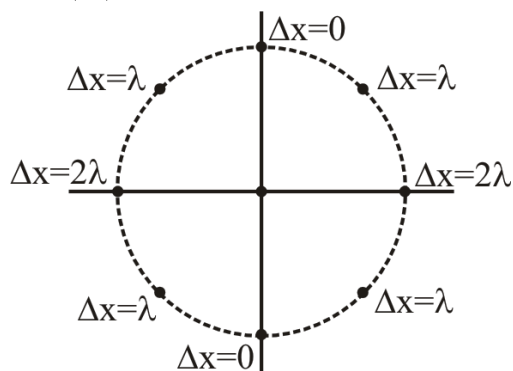
$$T = 12 \text{ sec.}$$

$$t_x = 2(\text{second time})$$

$$= \frac{T}{4} + \frac{T}{4} + \frac{T}{6}$$

$$= \frac{8T}{12} = 8 \text{ sec.}$$

8. **Ans (8)**



9. **Ans (60)**

$$\gamma = \alpha_x + \alpha_y + \alpha_z$$

$$= 5 \times 10^{-5} + 5 \times 10^{-6} + 5 \times 10^{-6}$$

$$= (50 + 5 + 5) \times 10^{-6}$$

$$\gamma = 60 \times 10^{-6}$$

$$C = 60.$$

10. **Ans (6)**

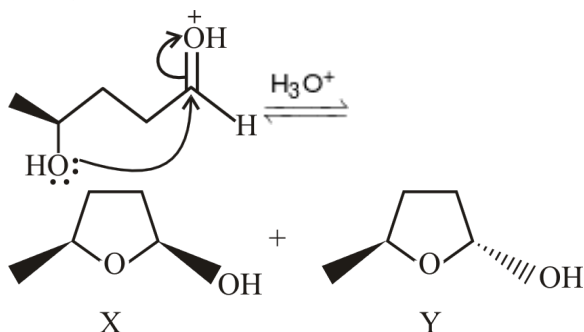
Magnetic field cannot change the speed of charged particle.

$$\text{So distance travelled} = vt = 2 \times 3 = 6m$$

PART-2 : CHEMISTRY

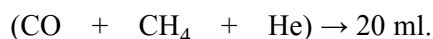
SECTION-I

2. Ans (C)

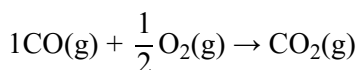


Products are diastereomer

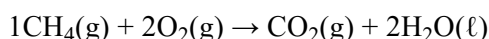
8. Ans (B)



$$a \text{ ml} \quad b \text{ ml} \quad 20 - (a+b) \text{ ml}$$



$$a \text{ ml} \quad a \text{ ml}$$



$$b \text{ ml} \quad b \text{ ml}$$

Contraction in vol = 13 ml

$$\text{or, } a\left(1 + \frac{1}{2} - 1\right) + b(1 + 2 - 1) = 13$$

$$\text{or, } \frac{a}{2} + 2b = 13$$

$$\text{or, } a + 4b = 26 \quad \dots\dots\dots(1)$$

$\therefore$ KOH absorbs CO_2 (a + b) ml

$$\therefore (a + b) = 14 \quad \dots\dots\dots(2)$$

$$(1) - (2) \Rightarrow 3b = 12$$

$$\boxed{b = 4 \text{ ml}}$$

$$\therefore \boxed{a = 10 \text{ ml}}$$

$$\therefore \text{CO} = 10 \text{ ml} \Rightarrow \frac{10}{20} \times 100 = 50\%$$

$$\text{CH}_4 = 4 \text{ ml} \Rightarrow \frac{4}{20} \times 100 = 20\%$$

$$\text{He} = 20 - (10 + 4) = 6 \text{ ml.}$$

$$\Rightarrow \frac{6}{20} \times 100 = 30\%$$

9. Ans (C)

$$\Lambda_m^{\circ} \text{AgCl} = \Lambda_m^{\circ} \text{AgNO}_3 + \Lambda_m^{\circ} \text{NaCl} - \Lambda_m^{\circ} \text{NaNO}_3$$

$$= 117 + 111 - 104 = 124 \text{ S cm}^2 \text{ mol}^{-1}$$

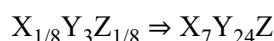
$$\Lambda_m^{\circ} \text{AgCl} = \frac{(K_{\text{AgCl sol.}} - K_{\text{H}_2\text{O}}) \times 1000}{S}$$

$$\text{or, } 124 = \frac{(2.4 \times 10^{-6} - 1.16 \times 10^{-6}) \times 1000}{S}$$

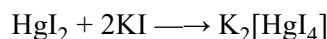
$$\text{or, } S = \frac{1.24 \times 10^{-3}}{124} = 10^{-5} \text{ mole/L}$$

$$K_{\text{SP}} \text{AgCl} = S^2 = (10^{-5})^2 = \boxed{10^{-10}}$$

10. Ans (B)



11. Ans (A)



$\Rightarrow$ No. of particles decreases $\Rightarrow$ F.P. is raised.

12. Ans (C)

$$n_A = 3 \Rightarrow \frac{t}{(t_{1/2})_A} = 3 \Rightarrow t = 3 \times 0.231 = 0.693$$

$$n_B = \frac{t}{(t_{1/2})_B} = \frac{0.693}{\left(\frac{0.693}{2}\right)} = 2$$

$$\frac{C_A}{C_{A_0}} = \left(\frac{1}{2}\right)^3 \quad \dots\dots\dots(1)$$

$$\frac{C_B}{C_{B_0}} = \left(\frac{1}{2}\right)^2 \quad \dots\dots\dots(2)$$

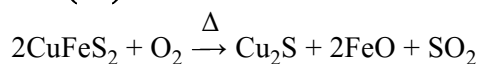
$$\frac{(1)}{(2)} \Rightarrow \frac{C_A}{C_{A_0}} \times \frac{C_{B_0}}{C_B} = \frac{1}{2}$$

$$\Rightarrow \frac{C_A}{C_B} = \frac{C_{A_0}}{C_{B_0}} \times \frac{1}{2} = \frac{3}{2} \times \frac{1}{2} = \boxed{\frac{3}{4}}$$

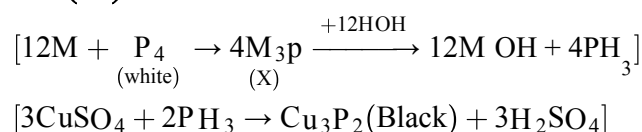
13. Ans (B)

Gold sol is an example of multimolecular colloidal particles.

15. Ans (C)



16. Ans (C)



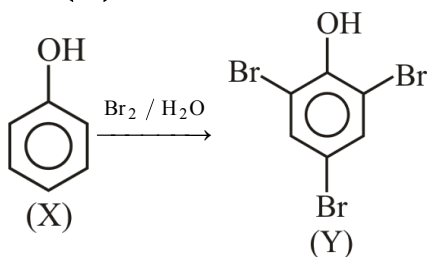
19. Ans (B)

Strength of O-D-bond > O-H bond

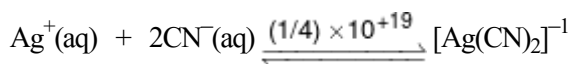
PART-2 : CHEMISTRY

SECTION-II

2. Ans (3)



4. Ans (75)



Initial conc. 0.03 M 0.1 M 0

Conc. at eq. 0 0.1 - 2 × 0.03 0.03

$$= 0.04$$

x 0.04 + 2x 0.03 - x

$$\approx 0.04 \approx 0.03$$

$$\frac{10^{19}}{4} = \frac{[\text{Ag}(\text{CN})_2]^{-1}}{[\text{Ag}^+][\text{CN}^-]^2} \text{ or } \frac{10^{19}}{4} = \frac{0.03}{[\text{Ag}^+] \times (0.04)^2}$$

$$\Rightarrow [\text{Ag}^+] = 7.5 \times 10^{-18}$$

$$\therefore \boxed{x = 75}$$

5. Ans (1200)

$$W = -\frac{1}{2}(P_1 + P_2)(V_2 - V_1) = -\frac{1}{2}(5 + 3)(4 - 1)$$

$$= -12 \text{ bar L}$$

$$\therefore \text{Work done by the system} = 12 \text{ bar L} = 1200 \text{ J.}$$

6. Ans (6)

$$V = 2 + 2 = 4 \text{ volt.}$$

$$\lambda = \sqrt{\frac{150}{V}} = \sqrt{\frac{150}{4}} = \sqrt{37.4} \text{ Å} = 6.12 \text{ Å}$$

$$\text{Ans.} = \boxed{6}$$

7. Ans (4)

$$D \propto \lambda \times u_{\text{avg}} \quad D = \text{diffusion coefficient}$$

$$\text{or, } D \propto \frac{T}{P} \times \sqrt{T}$$

$$\text{or, } D \propto \frac{T^{3/2}}{P} \dots\dots (1)$$

$$\Rightarrow \frac{D_1}{D_2} = \left(\frac{T_1}{T_2}\right)^{3/2} \times \frac{P_2}{P_1}$$

$$\text{or } \frac{D_1}{D_2} = \left(\frac{1}{4}\right)^{3/2} \times 2$$

$$\Rightarrow \frac{D_1}{D_2} = \left(\frac{1}{2}\right)^{3 \times \frac{3}{2}} \times 2 = \frac{1}{8} \times 2 = \frac{1}{4}$$

$$\text{Clearly } \boxed{D_2 = 4D_1} \Rightarrow \boxed{x = 4}$$

$$T_2 = 4T_1 \Rightarrow \frac{T_2}{T_1} = 4$$

$$P_2 = 2P_1 \Rightarrow \frac{P_2}{P_1} = 2$$

$$\lambda = \frac{KT}{\sqrt{2\pi}^2 P} \Rightarrow \lambda \propto \frac{T}{P}$$

$$u_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$$

$$u_{\text{avg}} \propto \sqrt{T}$$

$$\text{Ans.} = \boxed{4}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)

$$(3 + \cos 4x) \cos 2x = \cos 2x (3 + 2\cos^2 2x - 1)$$

$$= 2(\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x)[1 + \cos^2 2x]$$

$$= 2(\cos^4 x - \sin^4 x)[(\sin^2 x + \cos^2 x)^2 + (\cos^2 x - \sin^2 x)^2]$$

$$= 4(\cos^8 x - \sin^8 x)$$

$$\therefore \sum_{x=1^\circ}^{89^\circ} (3 + \cos 4x) \cos 2x = \sum_{x=1^\circ}^{89^\circ} 4[\cos^8 x - \cos^8 (90^\circ - x)]$$

$$= 4 \left(\sum_{x=1^\circ}^{89^\circ} \cos^8 x - \sum_{x=1^\circ}^{89^\circ} \cos^8 x \right) = 0$$

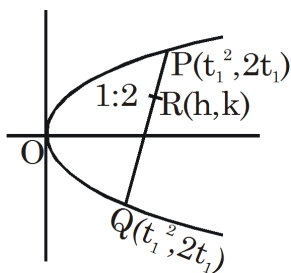
2. **Ans (C)**

$$\frac{2t_1 - 2t_2}{t_1^2 - t_2^2} = 2$$

$$\Rightarrow t_1 + t_2 = 1 \quad \dots(i)$$

$$h = \frac{2t_1^2 + t_2^2}{3} \quad \dots(ii)$$

$$k = \frac{4t_1 + 2t_2}{3} \quad \dots(iii)$$



$\therefore$ eliminating t_1, t_2 from (i), (ii) and (iii) and

replacing $h \rightarrow x$ and $k \rightarrow y$

$$\left(y - \frac{8}{9}\right)^2 = \frac{4}{9} \left(x - \frac{2}{9}\right)$$

3. **Ans (C)**

Let $x + 2 = t$ then $dx = dt$

$$\therefore I = \int \frac{x+1}{\{(x+2)^2 + 1\}} dx = \int \frac{t-1}{(t^2 + 1)^2} dt$$

$$dt = \int \frac{t}{(t^2 + 1)^2} - \frac{1}{(t^2 + 1)^2} dt$$

$$= \frac{-1}{2} \cdot \frac{1}{1+t^2} - \frac{1}{2} \int \frac{(1+t^2) - (t^2 - 1)}{(t^2 + 1)^2} dt$$

$$dt = \frac{-1}{2} \cdot \frac{1}{1+t^2} - \frac{1}{2} \tan^{-1} t + \frac{1}{2} \int \frac{\left(1 - \frac{1}{t^2}\right)}{\left(t + \frac{1}{t}\right)^2} dt$$

$$= \frac{-1}{2(1+t^2)} - \frac{1}{2} \cdot \tan^{-1} t + \frac{-1}{2\left(+\frac{1}{t}\right)} + c$$

$$= \frac{-1}{2(1+t^2)} - \frac{1}{2} \tan^{-1} t - \frac{t}{2(t^2 + 1)} + c$$

$$= -\frac{1+t}{2(1+t^2)} - \frac{1}{2} \tan^{-1} t + c$$

$$= -\frac{x+3}{2(x^2 + 4x + 5)} - \frac{1}{2} \tan^{-1}(x+2) + c$$

$$\therefore f(0) = -\frac{3}{10} \text{ and } g(-1) = -\frac{\pi}{8}$$

$$\therefore f(x) = \frac{-(x+3)}{2(x^2 + 4x + 5)} \text{ and } g(x) = -\frac{1}{2} \tan^{-1}(x+2)$$

$$\therefore f(1) + g(\sqrt{3} - 2) = -\frac{2}{10} - \frac{1}{2} \tan^{-1}(\sqrt{3}) = -\frac{1}{5} - \frac{\pi}{6}$$

4. **Ans (B)**

$\sim p$	p	q	$p \rightarrow q$	$q \vee \sim p$	$(p \rightarrow q) \leftrightarrow q \vee \sim p$
F	T	T	T	T	T
F	T	F	F	F	T
T	F	T	T	T	T
T	F	F	T	T	T

Hence tautology

5. **Ans (A)**

$$f(0) = \lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x \cos 3x}{\sin^2 2x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x \cos 3x}{4x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x \cos 2x \cos 3x + 2 \cos x \sin 2x \cos 3x + 3 \cos x \cos 2x \sin 3x}{8x}$$

(Use L'Hopital Rule)

$$= \frac{1 + 2^2 + 3^2}{8} = \frac{14}{8} = \frac{7}{4}$$

6. **Ans (C)**

If α, β, γ are roots of the equation $x^3 + qx + q = 0$ then

$$\Rightarrow (\alpha + \beta)^{-1} + (\beta + \gamma)^{-1} + (\gamma + \alpha)^{-1}$$

$$\Rightarrow -\frac{1}{\gamma} - \frac{1}{\alpha} - \frac{1}{\beta} \Rightarrow -\left(\frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma}\right)$$

$$\{\alpha\beta + \beta\gamma + \gamma\alpha = q, \alpha\beta\gamma = -q\}$$

$$\Rightarrow 1$$

7. **Ans (D)**

$$I_2 = \int_0^1 \frac{x^2 dx}{e^{x^3} (2 - x^3)}$$

$$\text{put } x^3 = t \Rightarrow 3x^2 dx = dt$$

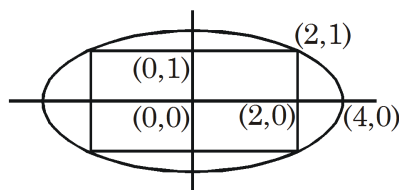
$$I_2 = \frac{1}{3} \int_0^1 \frac{dt}{e^t (2 - t)}$$

$$\text{put } 1 - t = z \Rightarrow dt = -dz$$

$$I_2 = \frac{1}{3} \int_1^0 \frac{-dz}{e^{1-z} (1+z)} = \frac{1}{3e} \int_0^1 \frac{e^z dz}{(1+z)} = \frac{I_1}{3e}$$

$$\Rightarrow \frac{I_1}{I_2} = 3e$$

8. **Ans (B)**



Let equation be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{16}{a^2} + 0 = 1 \quad \therefore a^2 = 16$$

$$\text{and } \frac{4}{a^2} + \frac{1}{b^2} = 1$$

$$\frac{1}{b^2} = \frac{3}{4}$$

$$b^2 = \frac{4}{3}$$

$$\frac{x^2}{16} + \frac{y^2}{4/3} = 1$$

$$e = \sqrt{1 - \frac{4}{3 \times 16}} = \sqrt{\frac{11}{12}} = \sqrt{\frac{11}{2\sqrt{3}}}$$

9. **Ans (B)**

$$f'(x) = 3x^2 + a$$

$$\Rightarrow f'(a) = f'(b) \Rightarrow 3a^2 + a = 3b^2 + a$$

$$\Rightarrow a = \pm b$$

as $a \neq b$ so only $a = -b$

$$\text{Now } f(1) = 1 + a + b = 1 \text{ (as } a = -b)$$

10. **Ans (C)**

Since, $(1, 2) \in S$ but $(2, 1) \notin S$

$\therefore S$ is not symmetric

Hence, S is not equivalence relation.

Given, $T = \{(x, y) : x - y \in I\}$

Now, $x - x = 0 \in I$, it is reflexive relation.

$$x - y \in I$$

$\Rightarrow y - x \in I$, it is symmetric relation.

$$\text{Let } x - y = I_1 \text{ and } y - z = I_2$$

$$\text{Now, } x - z = (x - y) + (y - z) = I_1 + I_2 \in I$$

$\therefore T$ is also transitive.

Hence, T is an equivalence relation.

11. **Ans (D)**

$$\text{Sum of roots} = \text{tr}(A) = 5$$

$$\text{product of roots} = |A|$$

$$\text{Characteristic equation is } x^2 - 5x + |A|$$

$$\text{since } f(A) = A^2 + aA + 3I$$

$$\therefore a = -5, \text{ and } |A| = 3$$

$$\text{consider } A = \begin{bmatrix} p & s \\ r & q \end{bmatrix} \text{ where } p + q = 5, pq - rs = 3$$

$$\text{adj}A = \begin{bmatrix} q & -s \\ -r & p \end{bmatrix}$$

$$(\text{adj}A)^2 = \begin{bmatrix} q & -s \\ -r & p \end{bmatrix} \begin{bmatrix} q & -s \\ -r & p \end{bmatrix}$$

$$= \begin{bmatrix} q^2 + rs & -s(p + q) \\ -r(p + q) & p^2 + rs \end{bmatrix}$$

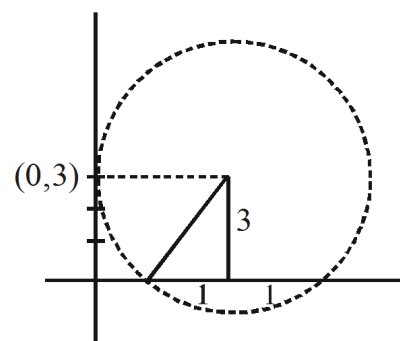
$$\text{tr}((\text{adj}A)^2 + (a + 1)\text{adj}A + 3I)$$

$$= p^2 + q^2 + 2rs - 4[p + q] + 3(1 + 1)$$

$$(p + q)^2 - 2(pq - rs) - 4(p + q) + 6$$

$$\Rightarrow 25 - 6 - 20 + 6 = 5$$

12. **Ans (C)**



$$R = \text{Radius} = \sqrt{10} \text{ centre } (\sqrt{10}, 3)$$

$S = \text{Equation of circle}$

$$= (x - \sqrt{10})^2 + (y - 3)^2 = 10$$

Intercept made by circle $S = 0$ on line

$$\sqrt{10}x - 3y = 1$$

$$\text{Perpendicular distance } P = \left| \frac{10 - 9 - 1}{\sqrt{19}} \right| = 0$$

$$\text{Length of intercept} = 2\sqrt{R^2 - P^2}$$

$$= 2R$$

$$= 2\sqrt{10}$$

13. **Ans (D)**

Multiply R_1 by a , R_2 by b & R_3 by c & divide the determinant by abc .

Now take a , b & c common from c_1 , c_2 & c_3 .

Now use $C_1 \rightarrow C_1 + C_2 + C_3$ to get

$$(a^2 + b^2 + c^2 + 1) \begin{vmatrix} 1 & 1 & 1 \\ b^2 & b^2 + 1 & b^2 \\ c^2 & c^2 & c^2 + 1 \end{vmatrix} = 1.$$

Now use $c_1 \rightarrow c_1 - c_2$ & $c_2 \rightarrow c_2 - c_3$

we get $1 + a^2 + b^2 + c^2 = 1 \Rightarrow a = b = c = 0$

14. **Ans (D)**

$$z = x + iy \text{ and } x = |x - 2 + iy|$$

$$x^2 = (x - 2)^2 + y^2$$

$$4x = y^2 + 4$$

$$\text{Hence } 4x_1 = y_1^2 + 4 \quad \dots(i)$$

$$4x_2 = y_2^2 + 4 \quad \dots(ii)$$

$$\text{Also } \arg(x_1 - x_2) + i(y_1 - y_2) = \frac{\pi}{3}$$

$$\frac{y_1 - y_2}{x_1 - x_2} = \sqrt{3}$$

$$y_1 - y_2 = \sqrt{3}(x_1 - x_2) \quad \dots(iii)$$

$$\text{on solving (i), (ii), (iii) } y_1 + y_2 = \frac{4}{\sqrt{3}}$$

15. **Ans (C)**

If each observation is increased by a same quantity the variance remains unchanged.

$$\therefore \sigma_A : \sigma_B : \sigma_C = 1 : 1 : 1$$

16. **Ans (D)**

The given expression

$$= \frac{(1+x)^{101} - 1}{(1+x) - 1} = \frac{(1+x)^{101} - 1}{x}$$

The coefficient of x^7 in $\frac{(1+x)^{101} - 1}{x}$ is

equivalent to coefficient of x^8 in

$$(1+x)^{101} - 1 = {}^{101}C_8 = {}^{101}C_{93}$$

17. **Ans (B)**

$$\text{we have, } \lim_{n \rightarrow \infty} \sum_{r=1}^n \cot^{-1} \left(2^{r+1} + \frac{1}{2^r} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \cot^{-1} \left(\frac{2^{2r+1} + 1}{2^r} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \left(\frac{2^{r+1} - 2^r}{1 + 2^{r+1} \cdot 2^r} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n (\tan^{-1} (2^{r+1}) - \tan^{-1} (2^r))$$

$$= \lim_{n \rightarrow \infty} (\tan^{-1} (2^{n+1}) - \tan^{-1} 2)$$

$$= \frac{\pi}{2} - \tan^{-1} 2 = \cot^{-1} 2$$

18. **Ans (B)**

$$2(\sin \theta + 1)^2 = \sin \theta (6 \sin \theta + 6)$$

$$2 \sin \theta + 2 = 6 \sin \theta$$

$$\sin \theta + 1 = 3 \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

19. **Ans (B)**

$$\text{Let equation of plane is } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1,$$

$$\text{where } A \equiv (a, 0, 0), B \equiv (0, b, 0), C \equiv (0, 0, c)$$

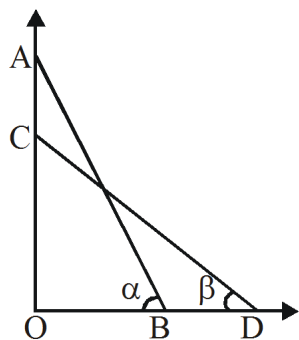
plane passes through $(1, -2, 3)$

$$\therefore \frac{1}{a} - \frac{2}{b} + \frac{3}{c} = 1$$

hence locus of point (a, b, c) is

$$yz - 2xz + 3xy = xyz$$

20. Ans (B)



Let length of ladder is $= \ell \therefore AB = CD = \ell$

$AC = 3$, $BD = 2$

In $\triangle AOB$ $AO = \ell \sin \alpha$, $OB = \ell \cos \alpha$

In $\triangle COD$ $CO = \ell \sin \beta$, $OD = \ell \cos \beta$

$$\frac{AC}{BD} = \frac{AO - CO}{OD - OB} = \frac{\sin \alpha - \sin \beta}{\cos \beta - \cos \alpha}$$

$$\frac{AC}{BD} = \frac{2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)}{2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)} = \cot \left(\frac{\alpha + \beta}{2} \right)$$

$$\therefore \tan \left(\frac{\alpha + \beta}{2} \right) = \frac{BD}{AC} = \frac{2}{3}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (112)

$P(x) - x^4$ is divisible by $(x-1)$, $(x-2)$, $(x-3)$, $(x-4)$ and $(x-5)$

$\therefore P(x) = k(x-1)(x-2)(x-3)(x-4)(x-5) + x^4$ where $k \in \mathbb{R} \sim \{0\}$

$\therefore P(0) = -120 \therefore k = 1$

$\therefore P(x) = (x-1)(x-2)(x-3)(x-4)(x-5) + x^4$

$\therefore P(x) = (y-2)(y-1)y(y+1)(y+2) + (y+3)^4$

where $y = x - 3$

$\therefore P(x) = y(y^4 - 5y^2 + 4) + (y+3)^4$

$P(x) = (x-3)^5 - 5(x-3)^3 + 4(x-3) + x^4$

$\therefore P'(x) = 5(x-3)^4 - 15(x-3)^2 + 4 + 4x^3$

$\therefore P'(3) = 0 + 4 + 4 \cdot 3^3 = 112$

2. Ans (75)

$$y' + \frac{1}{x}y = y^{\frac{1}{2}}$$

$$\Rightarrow y^{\frac{1}{2}}y' + \frac{1}{x}y^{\frac{1}{2}} = 1$$

$$\text{Put } y^{\frac{1}{2}} = v \Rightarrow v' = \frac{1}{2}y^{-\frac{1}{2}}y'$$

$$2v' + \frac{1}{x}v = 1 \Rightarrow v' + \frac{1}{2x}v = \frac{1}{2}$$

$$\text{Here } P = \frac{1}{2x}, Q = \frac{1}{2}$$

$$\text{If } = e^{\int P dx} = e^{\int \frac{1}{2x} dx} = e^{\frac{1}{2} \ln |x|} = |x|^{\frac{1}{2}}$$

$$\text{So, } v \times \text{IF} = \int \text{IF} \times Q$$

$$v \times |x|^{\frac{1}{2}} = \int |x|^{\frac{1}{2}} \times \frac{1}{2} \Rightarrow y^{\frac{1}{2}} |x|^{\frac{1}{2}} |x|^{\frac{1}{2}} = \frac{2|x|^{\frac{3}{2}}}{3} \times \frac{1}{2} + c$$

$$\text{Now, } y(1) = 0 \Rightarrow 0 = \frac{1}{3} + c \Rightarrow c = -\frac{1}{3}$$

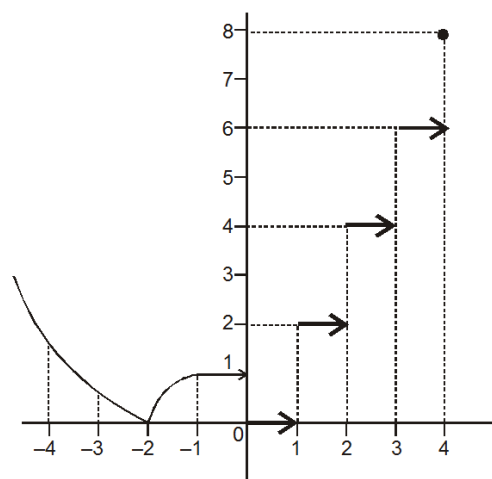
$$\therefore y = \left(\frac{1}{3}x - \frac{1}{3}x^{\frac{1}{2}} \right)^2 \Rightarrow y = \frac{x^3 - 2x^{\frac{3}{2}} + 1}{9x}$$

$$\Rightarrow y = \frac{1}{9} \left(x^2 - 2x^{\frac{1}{2}} + \frac{1}{x} \right)$$

$$\therefore [9f(9)] = 75$$

3. Ans (2)

Graph of $f(x)$ is



from fig. $f(x)$ is differentiable at $x = -3$, and $x = -1$ in $(-4, 4)$

4. Ans (10)

$$f'(x) = x^2 - (m-3)x + m \geq 0$$

$$D \leq 0$$

$$(m-3)^2 - 4m \leq 0$$

$$m \in [1, 9]$$

5. **Ans (21)**

A : coins shows no heads

B_1 : Die shows 1 or 2

B_2 : Die shows 3

B_3 : Die shows 4,5,6

$$P(B_1/A) = \frac{P(B_1) \cdot P(A/B_1)}{\sum_{i=1}^3 P(B_i) P(A/B_i)} = \frac{8}{13} = \frac{m}{n}$$

$$\therefore (n + m) = 21$$

6. **Ans (4)**

$$(\vec{r} - \vec{p}) \times (\vec{p} \times \vec{q}) = 0$$

$$\vec{r} - \vec{p} = \lambda (\vec{p} \times \vec{q})$$

$$\vec{r} = \vec{p} + \lambda (\vec{p} \times \vec{q})$$

$$(\vec{p} \times \vec{q}) \cdot \vec{r} = 12$$

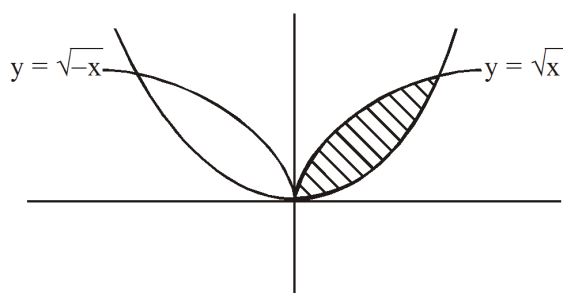
$$(\vec{p} \times \vec{q}) \cdot (\vec{p} + \lambda (\vec{p} \times \vec{q})) = 12$$

$$0 + \lambda |\vec{p} \times \vec{q}|^2 = 12$$

$$\lambda = \frac{4}{3}$$

$$|\vec{r} - \vec{p}| = 3\lambda = 4$$

7. **Ans (3)**



$$A = \left(\frac{16ab}{3} \right)$$

$$a = \frac{1}{4}; b = \frac{1}{4}$$

$$A = \frac{1}{3}$$

8. **Ans (2)**

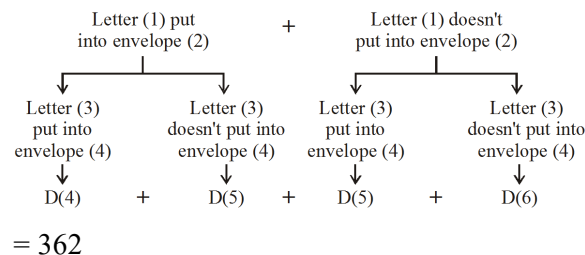
$$P = \frac{5}{\log_x 2 + \log_x 3 + \log_x 4 + \log_x 5} = \frac{5}{\log_x 120}$$

$$P = 5 \log_{120} x = \log_{120} x^5$$

$$(120)^P = 120^{\log_{120} x^5} = x^5$$

$$\text{so } x^5 = 32 \Rightarrow x = 2$$

9. **Ans (362)**



10. **Ans (9)**

$$y^{3/8} = k = 3^{\frac{1}{3}} \cdot 9^{\frac{1}{9}} \cdot 27^{\frac{1}{27}} \cdot 81^{\frac{1}{81}} \dots \infty$$

$$k = 3^{\frac{1}{3} \cdot 3 \cdot \frac{2}{9} \cdot 3 \cdot \frac{3}{27} \cdot 3 \cdot \frac{4}{81} \dots \infty}$$

$$k = 3^{\left(\frac{1}{3} + \frac{2}{9} + \frac{3}{27} + \frac{4}{81} + \dots \infty \right)}$$

$$\text{let } S = \frac{1}{3} + \frac{2}{9} + \frac{3}{27} + \frac{4}{81} + \dots \infty$$

$$\frac{1}{3}S = \frac{1}{9} + \frac{2}{27} + \frac{3}{81} + \dots \infty$$

$$\frac{2}{3}S = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \dots \infty$$

$$\frac{2}{3}S = \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{2} \Rightarrow S = \frac{3}{4}$$

$$\Rightarrow y^{\frac{3}{8}} = 3^{\frac{3}{4}} \Rightarrow y = \left(3^{\frac{3}{4}} \right)^{\frac{8}{3}} = 3^2 = 9$$